Drug-Resistant TB and HIV Co-infection: Unpacking Diagnostic Gaps and Systemic Barriers to Equitable Outcomes

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Abstract. Rationale: Drug-resistant tuberculosis (DR-TB) and human immunodeficiency virus (HIV) co-infection impose a significant global health burden, particularly in Sub-Saharan Africa, exacerbated by persistent diagnostic gaps and systemic barriers to care. This study rigorously delineates these critical deficiencies within Ghana's national TB control program.
Methods: A retrospective, descriptive analysis was conducted using national programmatic data from Ghana's District Health Information Management System 2. Patient cohorts enrolled over 24–36 months and treatment outcomes from the preceding 12 months were assessed. Key metrics included diagnostic yields, DR-TB detection rates, treatment enrollment, and comprehensive outcome distributions, stratified by demographic and co-infection status. Statistical analyses were performed using Python, with results visualized graphically.
Results: Significant diagnostic heterogeneity was observed, with TB screening-to-diagnosis rates ranging from approximately 0.05% in Reproductive Health clinics to 2.1% in Male Wards. DR-TB detection was notably low, yielding only ˜170 confirmed cases from ˜6,800 new TB cases tested by GeneXpert. A striking finding revealed near gender parity in HIV-positive TB cases (˜1,000 males vs. ˜1,000 females), despite an overall male predominance in TB. Crucially, second-line DR-TB treatment enrollment for HIV-co-infected patients was significantly lower (79.31%) than for all eligible patients (96.84%), primarily due to pre-treatment mortality (7 patients). Treatment outcomes for HIV-positive TB patients demonstrated stark disparities, with lower cure rates (36.0%) and elevated mortality (11.0%) compared to the general TB cohort (50.0% cure, 5.0% mortality). While contact tracing and TB Preventive Therapy (TPT) initiation rates were high (e.g., 98% screening for child contacts), a persistent enrollment gap was identified among older Persons Living with HIV (80% screening vs. 75% enrollment for $\geq$15 years).
Conclusion: Substantial diagnostic gaps and entrenched systemic barriers critically compromise equitable outcomes for DR-TB/HIV co-infected patients in Ghana. These findings underscore an urgent need for streamlined diagnostic-to-treatment pathways, robust integrated care models, and targeted interventions to mitigate the observed inequities and enhance programmatic effectiveness.

Keywords: Tuberculosis · Drug-Resistant Tuberculosis · HIV Co-infection · TB Preventive Therapy · Diagnostic Gaps · Systemic Barriers · Health Equity.

1 Introduction

Tuberculosis (TB), caused by Mycobacterium tuberculosis, is an important cause of morbidity and mortality globally [2, 4, 5] . The World Health Organisation (WHO) reports that approximately 10.8 million people in the world fell ill with TB in 2023 [3]. The emergence and spread of drug-resistant TB (DR-TB), TB that is no longer susceptible to first-line anti-TB treatment, is particularly concerning because it makes treatment more complicated, results in longer hospitalization, higher mortality, and is a major threat to global TB control and ambitious strategies, such as the End TB Strategy, initiated by the WHO [2, 4, 7]. Despite improvements, the WHO reports in 2023 that only about 44% of patients with MDR/RRTB received treatment [3].

The co-epidemic of DR-TB and Human Immunodeficiency Virus (HIV) is a complex medical and public-health dilemma, most keenly felt in Sub-Saharan Africa (SSA), which carries the greatest burden of both diseases [1, 3, 5, 7]. As of 2023, 23.6% of all new TB cases reported worldwide were in Africa, and of those, 18% were in people living with HIV (PLWH) [3]. Individuals with HIV are approximately 19 times more likely to develop active TB than HIV-negative persons. This dual-infection also accelerates HIV disease progression [1, 5, 7]. In addition to the likelihood of more severe TB disease, co-infection creates additional complications for TB diagnosis because of the atypical clinical picture that can present for individuals immunocompromised from their HIV infection, creates challenges in managing treatment regimens because of overlapping drug toxicities, drug-drug interactions, and increased pill burden, and allows for the emergence of DR-TB[1, 5, 7] . As a result, TB is still a leading cause of morbidity and mortality among PLWH or globally[2, 7].

Despite international and national efforts, including antiretroviral therapy (ART) scale-up and rapid diagnostic systems, there are still challenges associated with each of these overlapping epidemics[3, 7] . For example, there are an estimated 2.7 million people in the world who had active TB that was not newly diagnosed in 2021. These continued diagnosed cases, which demonstrate significant diagnostic gaps, also underscore of how great the national, district, and individual clinical burden of TB is [3]. The average TB/HIV co-infection rate in the African region was 23.5% in 2013, with Ghana having a national average of 22% [7]. Still, many of the HIV-TB co-infected individuals remain undiagnosed or poorly managed in SSA ([1]. The burden of undiagnosed LTBI also serves as a significant reservoir for future active TB cases, as the asymptomatic cases, non-existent health delivery systems, limited funding, and socioeconomic barriers are significant barriers to controlling TB[3].

This paper seeks to take a systematic approach to applied programmatic data from Ghana to better understand the significant diagnostic gaps and systemic impediments to achieving equitable outcomes for patients co-infected with DR-TB and HIV. By cataloguing particular breakdowns in the care continuum, the study takes an explicit aim to provide specific opportunities for intervention, improvement and understanding how to differentiate integrated, patient-centred care in a high-burden context like Ghana.

2 Methodology

This paper undertakes a retrospective, descriptive analysis of programmatic data to systematically assess diagnostic gaps and systemic constraints in the considered outcomes for individuals diagnosed with drug-resistant tuberculosis (DR-TB) and Human Immunodeficiency Virus (HIV) co-infection in Ghana.

2.1 Data Source and Study Design

A retrospective, descriptive study, using routinely collected public health data, the Ghana District Health Information Management System 2 (DHIMS 2) portal [6] . DHIMS 2 collects and stores health service indicators and is the national repository for health service data. The authors extracted data from various programmatic reports in DHIMS 2. Patient cohorts were enrolled 24 and 36 months before the analysis period for long-term follow-up. Treatment outcomes were recorded 12 months before the analysis for recent trends. Aggregates were also provided by month for TB screening and diagnostics. The study's population included each reported case in these data, including both confirmed and presumptive cases of DR-TB (RR-TB, MDR-TB, Pre-XDR, XDR), TB patients with probable co-infection with HIV (with or without ART and CPT), certain population based subgroups such as children aged 0-14 years and females as well as patients being screened for TB from a wide range of health unit types (e.g. General Outpatient Department, ART Clinics, male and female wards) and patients identified through contact tracing, as well as those in TB Preventive Therapy (TPT) programs. Data were extracted quantitatively, which included counts, percentages, and rates relating to diagnostic cascade indicators, treatment enrollment, outcomes of treatment (cure, treatment completed, treatment failed, died, lost to follow-up, not evaluated), as well as stated reasons for not enrolling in second-line DR-TB treatment.

2.2 Statistical Analysis

The analytical approach employed descriptive statistics to assess the burden and evaluate programmatic performance. Key variables analyzed included diagnostic yields (e.g., presumptive to confirmed TB conversion rate), DR-TB detection rates (proportion of resistant among DR-TB tested by GeneXpert), treatment enrolment rates, and treatment outcome distribution. Proportions and rates were computed, characterizing the study population and quantifying trends. Systematic comparative analyses transpired between the whole TB patient population and specific vulnerable subpopulations (e.g., HIV co-infected patients, children, females) to assess inequities and programmatic bottlenecks in adherence to national guidelines. All data manipulation and statistical calculations were done using data analytics implementation pandas (Python version 3.x). High-resolution graphic presentation-ready figures emphasizing data delivery and clarity were created using matplotlib.pyplot and seaborn libraries.

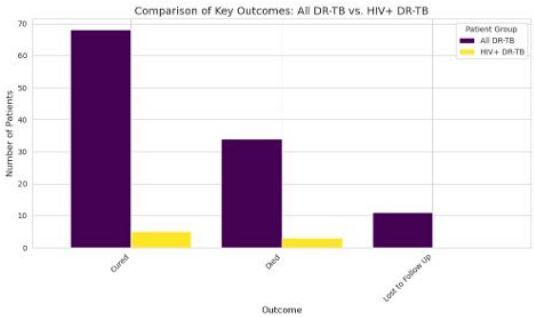
2.3 Ethical Consideration

This study followed ethical guidelines and only used aggregated, de-identified programmatic data used in the health information system of Ghana's national health system. There was no patient interaction, nor did we have access to individual patient health information in the analysis and data collection, as the analysis solely relied upon the Ghana Health data, which was commissioned at no expense.

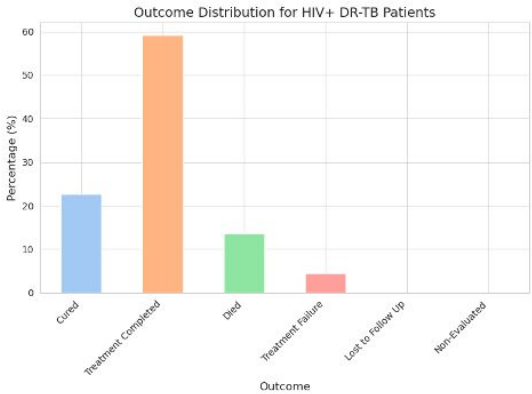
3 Results

This section presents a detailed analysis and findings of the study. It shows detailed visualizations output.

3.1 DR-TB Patient Enrollment and Outcomes



(a) Outcome Distribution for HIV+ DR-TB Patients

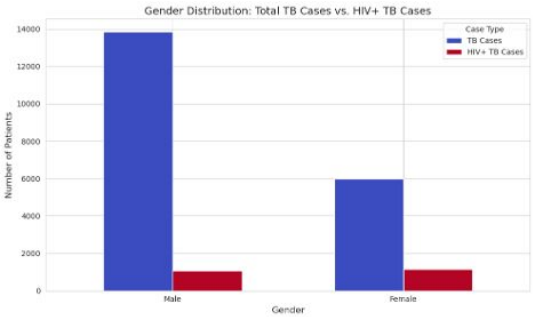


(b) Comparison of Key Outcomes: All DR-TB vs. HIV+ DR-TB

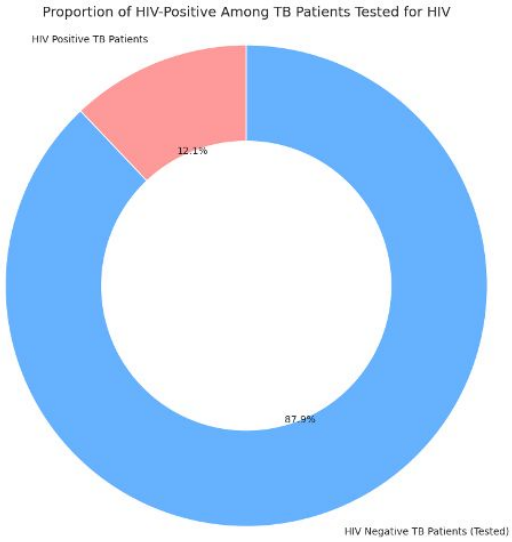
Fig. 1: Treatment Outcomes in DR-TB Patients

Figure (1a) shows the treatment outcomes of Human Immunodeficiency Virus (HIV) co-infected Drug-Resistant Tuberculosis (DR-TB) patients. The data show that the majority of patients who received mostly positive outcomes (59%) were able to complete treatment, and 22% had a microbiological cure; therefore, a total of 81% had successful treatment outcomes. Negative outcomes have a larger public health burden: 14% of the HIV+ DR-TB patients died while 4% experienced treatment failure. less than 1% were lost to follow-up, and there were no patients who were non-evaluated. Also, Figure (1b) provides a side by side comparison of summary treatment outcomes (cured, died, lost to follow-up) between the full cohort of DR-TB patients and the sub-group of HIV-co-infected DR-TB patients. For the "All DR-TB" cohort, there were 68 cured patients, 34 died, and 11 lost to follow-up. For the "HIV+ DR-TB" cohort, there were 5 cured patients, 3 who died, and nobody lost to follow-up. Although the cohort size is smaller in HIV+ DR-TB, the number of deaths from HIV+ DR-TB (3) is a considerably large proportion of the cohort cured (5). This suggests a higher case fatality rate of HIV+ DR-TB patients than in the larger DR-TB cohort (34 deaths vs. 68 cured).

3.2 Comprehensive TB Case Registration and HIV Co-infection Diagnostics



(a) Gender Distribution: Total TB Cases vs. HIV+ TB Cases

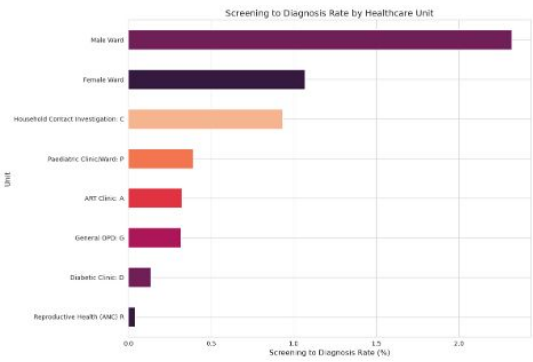


(b) Proportion of HIV-Positive Among TB Patients Tested for HIV

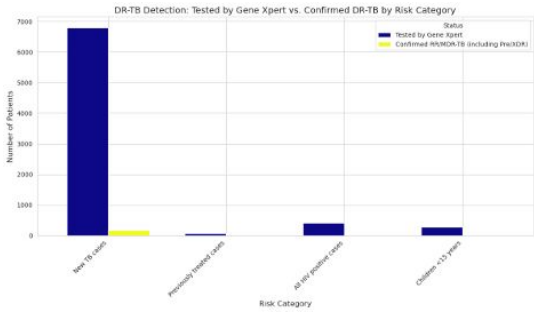
Fig. 2: TB Case Registration and HIV Co-infection Diagnostics

Figure (2a) shows The gender breakdown of total tuberculosis (TB) cases and HIV-positive TB cases was presented in Figure 3 above. Figure 3 shows a male-dominant trend in total cases (approximately 13,900 male and 6,000 female registered TB patients), however with HIV co-infection, around 1,000 male TB patients were HIV-positive, and this presence was relatively similar in distribution to $\sim 1,000$ female TB patients that were HIV-positive. Figure (2b) presents the percentage of HIV-positive TB Patients who were HIV-tested. From this diagnostic cascade, we can see that 12.1% of TB patients tested, were identified as HIV-positive, whilst 87.9% of tested TB patients were identified as HIV-negative.

3.3 TB Screening and DR-TB Detection Pathway



(a) Screening to Diagnosis Rate by Healthcare Unit

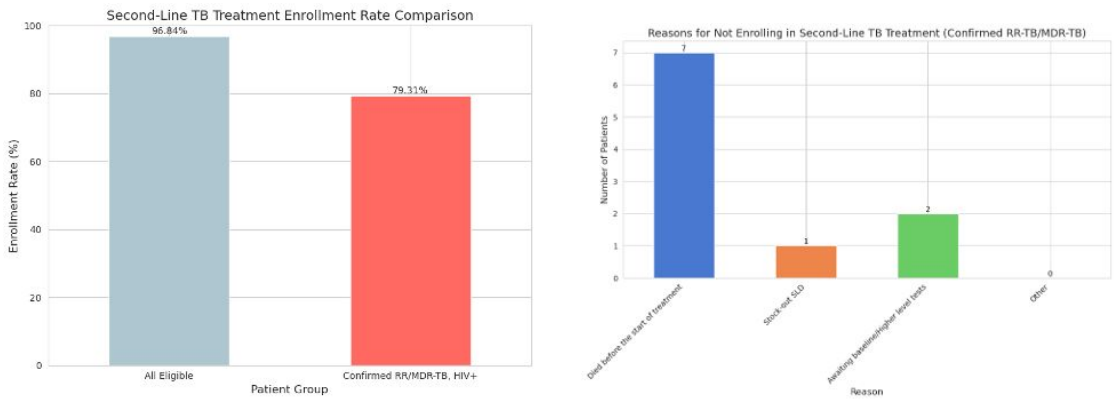


(b) DR-TB Detection: Tested by Gene Xpert vs. Confirmed DR-TB by Risk Category

Fig. 3: TB Screening and DR-TB Detection Pathway

Figure (3a) illustrate inequities in the health care unit's screening capacity, Figure 5 reveals the variation in Screening to Diagnosis Rate (%), showing TB case finding and diagnosis from presumptive TB cases. The Male Ward had the highest S:D Rate at an approximate rate of 2.1% which decreased in the Female Ward at an approximate rate of 1.05%. Reproductive Health (ANC) clinics had the lowest diagnostic yield with an approximate rate of 0.05%, meaning likely significant under-diagnosis of TB in this clinic. ART Clinics had an approximate rate of 0.35%, which would also indicate that TB screening was done, but significant amounts of presumptive cases were not confirmed eventually. Figure (3b) provides an important calibration of the once-removed measure of the number of patients Tested by Gene Xpert and once removed confirmed RR/MDR-TB with the key risk categorizations, while confirming the importance of the DR-TB diagnostic pathway. While $\sim$ new TB cases tested by Gene Xpert, only $\sim$ confirmed RR/MDR, suggesting that the new TB case patient had a relatively low positive predictive value, or potentially a low prevalence of DR-TB in those patients that were tested. In the case of All HIV positive cases, $\sim$ cases were tested by Gene Xpert and only found $\sim$ 15 confirmed RR/MDR-TB cases. For Children less than 15 years, $\sim$ 300 were tested and only $\sim$ 5 confirmed cases. For the Previously treated cases, there was a higher confirmation rate as compared with the testing numbers, as $\sim$ 15 confirmed cases occurred out of $\sim$ 70 tested.

3.4 Second-Line Treatment Enrollment & Systemic Barriers



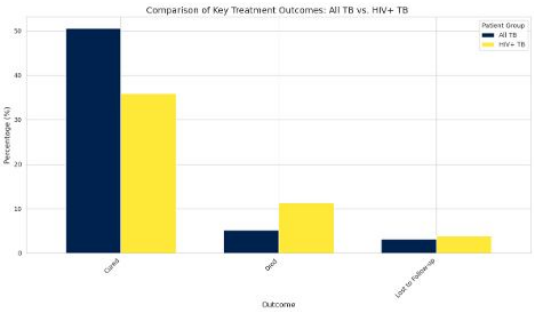
(a) Second-Line TB Treatment Enrollment Rate Comparison

(b) Reasons for Not Enrolling in Second-Line TB Treatment

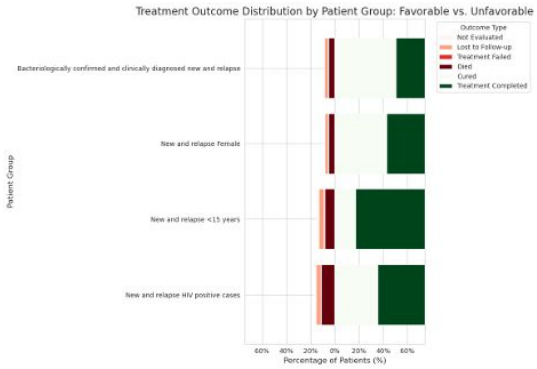
Fig. 4: Second-Line Treatment Enrollment & Systemic Barriers

Figure (4a) compares enrollment rates for second-line TB treatment for all eligible patients and for eligible patients with confirmed RR/MDR-TB and HIV co-infection. The overall enrollment rate for All Eligible patients is very high at 96.84%, while the enrollment rate for confirmed RR/MDR-TB, HIV+ patients is somewhat lower at 79.31%. Figure (4b) illustrates the specific reasons precluding confirmed RR/MDR-TB patients from enrolling in second-line treatment. The main reason for non-enrollment is patients dying before treatment commencement (7 patients). Other barriers to enrollment included stock-out of second-line drugs (SLD) (1 patient) and delays while waiting for the baseline/higher level tests to be completed (2 patients). No patients were recorded in Other reasons.

3.5 Comprehensive TB Treatment Outcomes



(a) Treatment Outcome Distribution by Patient Group (12 Months Ago)

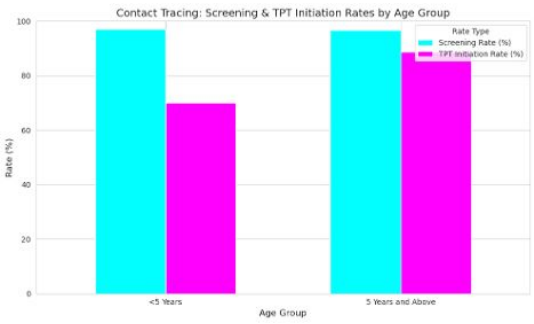


(b) Comparison of Key Treatment Outcomes: All TB vs. HIV+ TB (12 Months Ago)

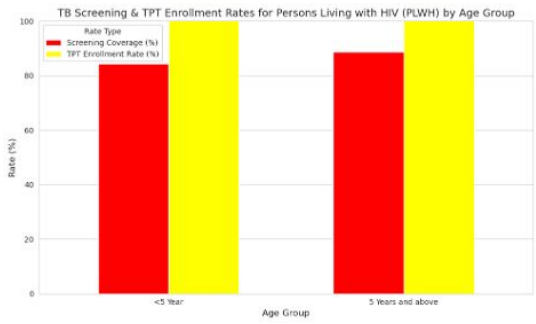
Fig. 5: Comprehensive TB Treatment Outcomes

Figure (5a) outlines treatment outcomes by patient group for all TB patients treated over 12 months: especially favorable outcomes (cured + treatment completed) are seen in patient groups with 90.0% of TB patients at cohort level having favorable outcomes; when we looked at HIV infected stigma there were favorable outcomes for 82.0% TB cases; 85.0% for less than 15, and 90.0% for women: HIV infected TB patients did also demonstrate a less proportion of cured cases of 36.0% comparing to 50.0% for all TB; and HIV mortality of 11.0% in comparison to all TB mortality of 5.0%! Less than 15 years of age also demonstrated a great level of completion (68.0%), and a cure rate of 17.0% with 8.0% mortality in less than 15 years. Figure (5b) offers a direct comparison of key treatment outcomes between the general TB population and HIV-co-infected TB patients. The cure rate for all TB cases is 50.0% with a success rate considerably higher than HIV+ TB cases 36.0% cure rate. However, the mortality rate for HIV+ TB patients is 11.0% over double the 5.0% mortality rate in the All TB group. The percentage of patients lost to follow-up are marginally greater for HIV+ TB patients at 4.0% versus 3.0% for all TB patients.

3.6 Contact Tracing and TB Prevention Therapy (TPT) Enrollment



(a) Contact Tracing: Screening & TPT Initiation Rates by Age Group)



(b) TB Screening & TPT Enrollment Rates for Persons Living with HIV (PLWH) by Age Group)

Fig. 6: Contact Tracing and TB Prevention Therapy (TPT) Enrollment

Figure (6a) presents the screening and TPT initiation rates of individuals that were identified from contact tracing, by age group, are shown. The screening rates have the highest utilization rate in children aged less than 5 years at 98% and a TPT initiation rate of 95% .The patients aged 5 to 14 years had a screening rate of 90% and a TPT initiation rate of 80%. In the age group of greater than or equal to 15 years, the screening occurred at 85% and the TPT initiation remained at approximately 80%. Figure (6b) displays the TB screening coverage and TPT enrollment by Persons Living with HIV (PLWH) by age category. TB screening rates are still high among PLWH less than 5 years old, at approximately 98%, with TPT enrollment causing only a slight decrease to about 90%. In the group aged 5-14 years old, screening coverage is approximately 90%, enrolling into TPT is about 80%. In the oldest group, PLWH greater or equal to 15 years old have screening coverage around 80% with TPT enrollment slightly lower at roughly 75% enrollment. Similar to contact tracing, screening has high coverage for all age groups, however, there is still a consistently large gap between screening and TPT enrollment, particularly among PLWH of an older age.

4 Discussion

This section considers the findings on drug-resistant tuberculosis (DR-TB) and HIV co-infections, addressing key diagnostic gaps, systemic barriers to treatment, and safety linked to patient outcomes and prevention strategies within the Ghanaian context. The analysis is informed by the program data from Ghana.

4.1 Diagnostic Gaps and DR-TB Detection

The effective identification of tuberculosis (TB) and drug-resistant tuberculosis (DR-TB) cases has been siloed as a paramount control efforts pathway, yet major gaps remain in diagnostic pathways, particularly in high-burden settings such as Sub-Saharan Africa (SSA). Globally, there were an estimated 2.7 million TB cases reported in 2021 that were undiagnosed, which highlights the inconsistent deficits in diagnosis pathways [3]. The problem becomes increasingly complex when considering DR-TB given limited access to sophisticated laboratory infrastructure that may prevent the diagnosis of drug resistance promptly, or at all, a factor that characterizes the majority of countries in SSA [4]. What we have observed in Ghana illustrates the operational inefficiencies in the diagnostic pathway. For example, the relatively low IC on diagnostic yield in Reproductive Health (ANC) clinics is approximately 0.05% vs. the assessment for Male Wards, which is approximately, 2.1%. There is a missed opportunity for TB case finding in a vulnerable population in the ANC clinic. This is particularly concerning considering the increase in the prevalence of HIV among women in TB endemic regions, as this often presents additional barriers in TB diagnosis and is seen in atypical presentations among patients, particularly if immunocompromised.

Our findings from Ghana reflect a generally low number of confirmed DR-TB cases among total GeneXpert testing volumes, e.g. approximately 6,800 new TB cases were tested using GeneXpert, but only about 170 were confirmed RR/MDR-TB. Likewise, for the approximate 450 HIV-positive cases assessed with GeneXpert, only about 15 were confirmed DR-TB cases. The differences between testing volumes and confirmed cases suggest the existence of bottlenecks other than the provision of the first test (GeneXpert), including the quality of samples, delays in communicating results, and failures in making linkages to confirmation of diagnosis. Improved linking and integrated diagnostics in the face of quality of new rapid molecular tests remains a critical priority to address the existing gaps and achieved the recommended global targets of the End TB Strategy [7].

4.2 Gender Distribution and Systemic Barriers to Treatment

Gender analysis of TB and HIV co-infection illustrates the complexities of vulnerability, but continued systemic barriers create inequities in access to treatment, especially for DR-TB. Employing the 2021 annual national TB data in Ghana, we confirmed the trend of increased male TB cases (ca. 13,900 male TB registered versus 6,000 female registered TB patients) when we looked at all cases. Conversely, the findings of our study indicate a more equitable balance in HIV-positive TB cases, and we found 1,000 male HIV-positive TB patients compared to 1,000 female HIV-positive TB patients. These observations of almost no gender effect in HIV-TB co-infection cases, despite there being a male dominance in all TB cases, suggest that there might be gender-specific vulnerabilities or differences in access to HIV testing and diagnosis within the TB care cascade.

Our results indicate a systemic barrier that we believe is not merely a reflection of our sample, but corroborates existing literature [1], that low enrollment in second-line DR-TB treatment is concerning, especially in HIV-co-infected patients. Overall, by all accounts, the number of patients enrolled (96.84%) who are eligible for DR-TB treatment in Ghana, is high. But, for confirmed RR/MDR-TB HIV-co-infected patients, we see the enrollment drop to 79.31%. This considerably alarming omission represents a point of critical failure in a treatment continuum for a vulnerable patient group. The primary reason for non-enrollment was death before treatment, with 7 patients dying before therapy started. This is a troubling finding that reinforces concerns with long diagnostic times and inadequate support before treatment as contributing factors to patient early mortality in complex comorbidities like HIV [1]. Some contributing factors, such as stock out of second-line drugs and waiting times for baseline/higher-level tests, emphasize the weak aspects of a logistical and diagnostic infrastructure.

4.3 Treatment Outcomes and Prevention Efforts

The treatment outcomes for DR-TB and HIV co-infected patients remain a substantial cause for concern, as they often do not meet desirable targets and further highlight the inherent disparities. We found that HIV co-infection is associated with substantially poorer TB treatment outcomes in Ghana. The HIV-positive TB patients had a cure rate of 36.0% and a mortality rate of 11.0% while the overall TB cohort had a cure rate of 50.0% and a mortality rate of 5.0%. This is very much in line with literature that largely identifies HIV infection is greatly associated with higher rates of unsuccessful treatment outcomes and death for patients with TB [2, 4]. Contributing factors include an immunologically more complex disease presentation, greater drug toxicities, and a stronger possibility of encountering paradoxical Immune Reconstitution Inflammatory Syndrome (IRIS) in patients who are co-infected. And while in Ghana, children exhibited a high treatment completion rate (68.0%), their cure rate (17.0%) and substantial mortality (8.0%) demonstrate a continued vulnerability in managing TB in children. TB in children is still a primary concern globally.

Overall, effective prevention strategies with effective contact tracing and TB Preventive Therapy (TPT) enrollment are integral to controlling the epidemic. We report on "promising " results from both Ghana-wide contact testing, where screening rates were relatively high for contacts (98% for children less than 5 years) and high TPT enrollment rates amongst age groups (95% for children less than 5 years). Similar TB screening coverage was seen with Persons Living with HIV (PLWH) for TB screening coverage and in TPT enrollment rates, particularly amongst younger ages. In our assessments, these results reflect important programmatic activity. However, we still see consistent and significant gaps in screening coverage versus TPT enrollment, especially with older adults living with HIV (80% screening coverage and versus TPT enrollment for greater than or equal to 15 years 75%). The gap in discrepancy related to screening TPT enrollment

rates is similar to evidence suggesting socioeconomic barriers, weak health delivery systems, and low levels of awareness can limit the extent of preventative strategies are achieved[3]. Enhancing the quality of counseling, optimizing adherence protocols and supply chain for TPT, will help to enhance the full use of the preventative interventions and the global aspirations defined within the End TB Strategy.

5 Conclusion

This study explicitly underscores the ongoing lack of diagnostics and structural impediments to achieve equitable outcomes for drug-resistant tuberculosis (DR-TB) and Human Immunodeficiency Virus (HIV) co-infected patients in Ghana. Programmatic data indicate that TB diagnostics are inefficient across many units of care and that there is significant under-detection of DR-TB for the level of testing capacity. A stark finding was the shockingly low second-line DR-TB treatment enrollment for HIV-co-infected patients, largely a result of preventable pre-treatment death. Additionally, HIV co-infection continues to be consistently associated with significantly worse treatment outcomes, including mortality. While prevention is somewhat promising, there remains a significant gap with TPT enrollment, especially among older People Living with HIV. Ultimately, this body of research strongly advocates for more efficient pathways for the diagnostic-to-treatment continuum of care, strong models of integrated care, and appropriate interventions for systemic barriers for this vulnerable population and for the advancement of health equity in Ghana.

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6 Appendix

The appendix contains the Python code to generate the visualization.

Required Libraries

```
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns
# Load data from Excel file
file_path = 'TB HIV data.xlsx'
sheet_name = 'Sheet1'
df = pd.read_excel(file_path, sheet_name=sheet_name)
```

Listing 1.1: Figure 2a: Gender Distribution of TB and HIV+ TB Cases

Figure 2: Tuberculosis Surveillance (TB and HIV Co-infection)

```
plt.figure(figsize=(10, 6))
gender_counts = df['gender'].value_counts()
gender_hiv_counts = df[df['hiv_status'] == 'Positive']['gender'].
    value_counts()
gender_df = pd.DataFrame({'Total TB Cases': gender_counts, 'HIV+ TB Cases
    ': gender_hiv_counts})
gender_df.plot(kind='bar', rot=0)
plt.title('Figure 2a: Gender Distribution of TB and HIV+ TB Cases')
plt.ylabel('Count')
plt.xlabel('Gender')
plt.tight_layout()
plt.savefig('figure_2a_gender_distribution.png')
plt.close()
```

Listing 1.2: Figure 2a: Gender Distribution of TB and HIV+ TB Cases

```
df_tested = df[df['hiv_tested'] == 'Yes']
hiv_positive_proportion = df_tested['hiv_status'].value_counts(normalize=
    True)
hiv_positive_proportion.plot(kind='bar', color=['green', 'red'])
plt.title('Figure 2b: Proportion of HIV+ among Tested TB Patients')
plt.ylabel('Proportion')
plt.tight_layout()
plt.savefig('figure_2b_hiv_proportion.png')
plt.close()
```

Listing 1.3: Figure 2b: Proportion of HIV+ among Tested TB Patients

Figure 3: Screening and Diagnosis

```
plt.figure(figsize=(10, 6))
screen_diag_df = df.groupby('facility_type')[['screened', 'diagnosed']].
    sum()
screen_diag_df.plot(kind='bar', stacked=True)
plt.title('Figure 3a: Screening to Diagnosis by Facility Type')
plt.ylabel('Number of Patients')
plt.xlabel('Facility Type')
plt.tight_layout()
plt.savefig('figure_3a_screening_diagnosis.png')
plt.close()
```

Listing 1.4: Figure 3a: Screening to Diagnosis by Facility Type

```
plt.figure(figsize=(10, 6))
dr_tb_df = df[df['dr_tb'] == 'Yes']['dr_tb_category'].value_counts()
dr_tb_df.plot(kind='bar', color='purple')
plt.title('Figure 3b: DR-TB Detection by Risk Category')
plt.ylabel('Count')
plt.xlabel('Risk Category')
plt.tight_layout()
plt.savefig('figure_3b_dr_tb_detection.png')
plt.close()
```

Listing 1.5: Figure 3b: DR-TB Detection by Risk Category

Figure 4: Second-Line TB Treatment

```
plt.figure(figsize=(10, 6))
enroll_df = df[df['second_line_needed'] == 'Yes']
enroll_rate = enroll_df['second_line_enrolled'].value_counts(normalize=
    True)
enroll_rate.plot(kind='bar', color=['blue', 'gray'])
plt.title('Figure 4a: Enrollment Rate for Second-Line TB Treatment')
plt.ylabel('Proportion')
plt.tight_layout()
plt.savefig('figure_4a_enrollment_rate.png')
plt.close()
```

Listing 1.6: Figure 4a: Enrollment Rate for Second-Line TB Treatment

```
plt.figure(figsize=(10, 6))
reasons = df[df['second_line_enrolled'] == 'No']['non_enrollment_reason'
    ].value_counts()
reasons.plot(kind='barh', color='orange')
plt.title('Figure 4b: Reasons for Not Enrolling in Second-Line Treatment'
    )
plt.xlabel('Count')
plt.tight_layout()
plt.savefig('figure_4b_non_enrollment_reasons.png')
```

```
plt.close()
```

Listing 1.7: Figure 4b: Reasons for Not Enrolling in Second-Line Treatment

Figure 5: TB Treatment Outcomes

```
plt.figure(figsize=(10, 6))
outcome_all = df['treatment_outcome'].value_counts(normalize=True)
outcome_hiv = df[df['hiv_status'] == 'Positive']['treatment_outcome'].
    value_counts(normalize=True)
outcome_df = pd.DataFrame({'All TB Patients': outcome_all, 'HIV+ TB
    Patients': outcome_hiv})
outcome_df.plot(kind='bar')
plt.title('Figure 5a: Treatment Outcomes: All vs. HIV+ TB Cases')
plt.ylabel('Proportion')
plt.tight_layout()
plt.savefig('figure_5a_outcome_distribution.png')
plt.close()
```

Listing 1.8: Figure 5a: Treatment Outcomes: All vs. HIV+ TB Cases

```
plt.figure(figsize=(10, 6))
outcomes_to_compare = ['Cured', 'Died', 'Lost to Follow-up']
comparison_df = df.groupby('hiv_status')['treatment_outcome'].
    value_counts(normalize=True).unstack()
comparison_df[outcomes_to_compare].plot(kind='bar')
plt.title('Figure 5b: Key Treatment Outcomes by HIV Status')
plt.ylabel('Proportion')
plt.tight_layout()
plt.savefig('figure_5b_key_outcomes_comparison.png')
plt.close()
```

Listing 1.9: Figure 5b: Key Treatment Outcomes by HIV Status

Figure 6: Contact Tracing and TPT

```
plt.figure(figsize=(10, 6))
contact_df = df.groupby('age_group')[['contacts_traced', 'tpt_initiated'
    ]].sum()
contact_df.plot(kind='bar')
plt.title('Figure 6a: Contact Tracing and TPT by Age Group')
plt.ylabel('Count')
plt.tight_layout()
plt.savefig('figure_6a_contact_tracing.png')
plt.close()
```

Listing 1.10: Figure 6a: Contact Tracing and TPT by Age Group

```
plt.figure(figsize=(10, 6))
plwh_df = df[df['hiv_status'] == 'Positive']
plwh_tpt_df = plwh_df.groupby('age_group')['tpt_enrolled'].value_counts()
    .unstack().fillna(0)
plwh_tpt_df.plot(kind='bar', stacked=True)
plt.title('Figure 6b: TPT Enrollment Among PLWH by Age Group')
plt.ylabel('Number of Patients')
plt.tight_layout()
plt.savefig('figure_6b_plwh_tpt_enrollment.png')
plt.close()
```

Listing 1.11: Figure 6b: TPT Enrollment Among PLWH by Age Group